

Aurora Research Organization

The Aurora Research Organization was established in 2022 to improve enterprise search systems. During the early stages of development, engineers relied entirely on keyword search. Users could retrieve documents only if they searched using the exact words present in the document. This caused many retrieval failures because different people often described the same concept using different words.

For example, one engineer searched for "backend services," while another document described the same work as "REST APIs." Although both referred to the same concept, keyword search failed to connect them. This problem motivated the research team to investigate semantic search.

Tourism Analytics Division

The Aurora Mountain Region is one of the most visited tourist destinations during winter. Every year thousands of tourists visit the mountains for trekking, photography, and snow activities. Local businesses depend heavily on tourism revenue.

Researchers also observed that tourists who visited nearby lakes spent more money than tourists who visited only the mountains.

Software Development Department

The software engineering department developed an e-commerce platform using Python, FastAPI, and MySQL.

The project included:

- Product Management APIs
- Authentication
- Authorization
- Payment Gateway Integration
- Order Tracking
- Customer Management

The application handled more than 10,000 users every day.

Database indexing significantly improved query performance.

Agriculture Research

Aurora Valley is famous for apple farming.

Researchers developed smart irrigation systems that monitor soil moisture using sensors.

Rainfall has become unpredictable during recent years.

The agriculture department expects sensor-based irrigation to reduce water consumption by twenty percent over the next five years.

Artificial Intelligence Research

After several months, the research team started studying Retrieval-Augmented Generation (RAG).

They learned that a RAG system consists of several components:

- Document Loader
- Chunking
- Embeddings
- Vector Database
- Retriever
- Large Language Model

The team realized that retrieval quality directly affects answer quality.

Banking Platform

The engineering team also developed a banking application.

The system maintained:

- Customer Accounts
- Balances
- Transactions
- Monthly Interest
- Audit Logs

Strict authentication and authorization protected customer information.

Embedding Research Part 1

The research team learned that embeddings convert text into vectors.

Unlike keyword search, embeddings represent the meaning of text.

Two completely different sentences may produce similar embeddings if they have similar meanings.

For example:

"Aziz built backend services."

and

"Aziz developed REST APIs."

These sentences have different words but almost identical meaning.

Travel Research

Visitors preferred traveling during winter because of pleasant weather.

Popular destinations included:

- Aurora Mountains
- Crystal Lake
- Green Valley

Hotels reported their highest occupancy during December and January.

Embedding Research Part 2

Engineers learned that cosine similarity measures how similar two vectors are.

If two vectors point in nearly the same direction, they are considered semantically similar.

This allows a search engine to retrieve information even when users use completely different wording.

Chunking Experiments

The engineering team compared three chunking strategies.

The first strategy used Recursive Chunking.

Recursive Chunking attempted to preserve paragraph boundaries and sentence boundaries whenever possible.

The second strategy used Token Chunking.

Instead of counting characters, Token Chunking counted LLM tokens.

This made it suitable for GPT-based models.

Finally, the researchers implemented Semantic Chunking.

Semantic Chunking grouped together sentences discussing the same topic.

Instead of splitting purely by size, it tried to preserve the meaning of the document.

Food Research

Researchers surveyed tourists about food preferences.

Popular foods included:

- Fish Curry
- Chicken Biryani
- Grilled Seafood
- Vegetable Curry

Tourists preferred spicy food during winter.

Restaurants expanded seasonal menus accordingly.

Hybrid Search Research

Although semantic search produced excellent results for natural language questions, researchers noticed one limitation.

Queries containing product codes, invoice numbers, customer IDs, or error codes often performed better with keyword search.

The engineering team therefore combined semantic search with keyword search.

This technique became known as Hybrid Search.

Hybrid Search executes both searches independently and merges the results before sending them to the language model.

Vector Database Evaluation

The team compared multiple vector databases.

FAISS performed very well for local development.

Pinecone simplified cloud deployments because it managed indexing automatically.

Both databases stored embeddings rather than the original text.

Final Research Summary

The Aurora Research Organization concluded that modern RAG systems depend on high-quality retrieval.

Recursive Chunking works well for general-purpose applications.

Token Chunking is useful when working with LLM context windows.

Semantic Chunking produces more meaningful chunks because it understands relationships between sentences.

Hybrid Search combines semantic retrieval and keyword retrieval to improve accuracy.

Embeddings convert text into vectors.

Cosine similarity compares vectors.

Vector databases store embeddings.

The retriever selects relevant chunks.

The language model generates the final answer.